

T11

$$H_0: \xi \sim p_0(x) = 1 \{0; 1\}$$

$$H_1: \xi \sim p_1(x) = \frac{e^{1-x}}{e-1} \{0; 1\}$$

a) $n=1$ сур. знач. d

$$l(x) = \frac{L_1}{L_0} = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \geq c$$

$$G: x \leq A$$

$$P(x \leq A | H_0)$$

$$\int_0^A 1 \cdot dx = d$$

$$A = d$$

$$G: x \leq d$$

?
если $x \leq d$
→ сюр. равномерно распределен

$$W = P(x \leq A | H_1) = \int_0^A \frac{e^{1-x}}{e-1} dx =$$

$$= \frac{1}{e-1} \int_0^A e^{1-x} dx = \frac{-1}{e-1} \int_0^A de^{1-x} = \frac{-1}{e-1} e^{1-x} \Big|_0^A =$$

$$= \frac{-1}{e-1} (e^{1-d} - e) = \frac{e}{e-1} (1 - e^{-d})$$

$$\alpha_1 = d$$

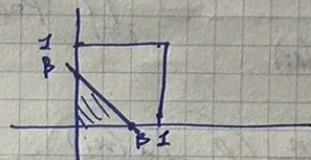
$$\alpha_2 = 1 - W$$

$$8) n=2$$

$$I = \frac{I_1}{I_0} = \frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq C$$

$$G_1: x_1 + x_2 \leq B$$

$$P(x_1 + x_2 \leq B | H_0) = \alpha$$



$$\iint_{x_1+x_2 \leq B} 1 \cdot 1 \, dx_1 \, dx_2 = \alpha$$

$$\frac{B^2}{2} \leq \alpha \quad B = \sqrt{2\alpha} \quad B \leq 1$$

$$\alpha \leq \frac{1}{2}$$

$$G: x_1 + x_2 \leq B = \sqrt{2\alpha}$$

$$e^{1-x} \Big|_0^A =$$

$$W = P(x_1 + x_2 \leq B | H_1) = \iint_{x_1+x_2 \leq B} \frac{e^{1-x_1} \cdot e^{1-x_2}}{(e-1)^2} \, dx_1 \, dx_2 =$$

$$= \frac{e^2}{(e-1)^2} \int_0^B dx_2 \int_0^{B-x_2} e^{-x_1} e^{-x_2} \, dx_1 = \left(\frac{e}{e-1}\right)^2 [1 - e^{-B} - e^{-B} B]$$

$$\alpha_1 = \alpha$$

$$\alpha_2 = 1 - W$$

c) ассимпт. кр. по выборке объема n

$$\begin{aligned} H_0: p_0(x) &= 1 \{0, 1\} \\ H_1: p_1(x) &= \frac{e^{1-x}}{e-1} \{0, 1\} \end{aligned} \quad \left| \quad \ell(\vec{x}_n) = \frac{\ell_1(\vec{x}_n)}{\ell_0(\vec{x}_n)} = \right.$$

$$= \frac{\prod_{i=1}^n p_1(x_i)}{\prod_{i=1}^n p_0(x_i)} = \prod_{i=1}^n \frac{p_1(x_i)}{p_0(x_i)} \geq C$$

$$\ln \ell = \sum_{i=1}^n \ln \frac{p_1(x_i)}{p_0(x_i)} \geq \ln C$$

$$\sum_{i=1}^n \ln p_1(x_i) - \sum_{i=1}^n \ln p_0(x_i) = \sum_{i=1}^n \ln \left(\frac{e^{1-x_i}}{e-1} \right) - 0 \geq \ln C$$

$$\sum_{i=1}^n (1-x_i) - n \ln(e-1) \geq \ln C$$

$$G: \sum_{i=1}^n x_i \leq A$$

$$\bar{x} \leq B$$

$$P(\bar{x} \leq B | H_0) = \alpha$$

н.м.т. (наименьшая)

$$H_0: M\xi = 1/2$$

$$H_1: \xi = \frac{1}{12}$$

$$P(\bar{x} \leq B | H_0) = P\left(\frac{\bar{x} - 1/2}{\sqrt{1/12}} \leq \frac{B - 1/2}{\sqrt{1/12}}\right)$$

$$B = \frac{1}{2} + \frac{U_\alpha}{\sqrt{12n}}$$

$$\frac{\bar{x} - M\xi}{\sqrt{D\xi}} \xrightarrow{\sim} N(0, 1) \quad G: \bar{x} \leq \frac{1}{2} + \frac{U_\alpha}{\sqrt{12n}}$$

$$W = P(\bar{x} \leq B | H_1)$$

$$H_1: M\xi = \int_0^1 x p_1(x) = \frac{e-2}{e-1} = 0,418$$

$$M\xi^2 = \frac{2e-5}{e-1}$$

$$D\xi = \frac{e^2 - 5e + 1}{(e-1)^2}$$

$$W = P \left(\frac{\bar{x} - M\bar{\xi} \sqrt{n}}{\sqrt{D\bar{\xi}}} \leq \frac{B - M\bar{\xi} \sqrt{n}}{\sqrt{D\bar{\xi}}} \right) =$$

$$= \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

Согласно нормального

$$W \rightarrow 1$$

при $n \rightarrow \infty$

$$\gamma = \frac{\frac{1}{2} + \frac{U_d}{\sqrt{12}} - 0,418}{\sqrt{D\bar{\xi}}} \sqrt{n} = \frac{\frac{U_d}{\sqrt{12}} + 0,082 \cdot \sqrt{n}}{\sqrt{D\bar{\xi}}} \xrightarrow{n \rightarrow \infty} +\infty$$

$$\Rightarrow W \rightarrow 1$$

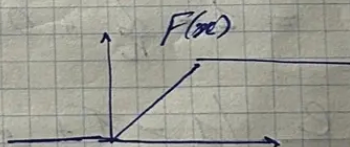
$$\alpha_1 = \alpha$$

$$\alpha_2 = 1 - \alpha$$

2) $X_{\min} \leq C$ — критерий отбора

$$\alpha_2 \rightarrow \min$$

$$\alpha_1 \leq \alpha$$



$$\alpha_1 = P(X_{\min} \leq C | H_0) = 1 - (1 - F(C))^n \leq \alpha$$

$$1 - (1 - C)^n \leq \alpha$$

$$-(1 - C)^n \leq \alpha - 1$$

$$(1 - C)^n \geq 1 - \alpha$$

$$1 - C \geq (1 - \alpha)^{1/n}$$

$$C \leq 1 - \sqrt[n]{1 - \alpha}$$

$$\alpha_2 = P(\alpha_{\min} > c | H_1) = 1 - P(\alpha_{\max} \leq c | H_1) =$$

$$= 1 - 1 + (1 - \Phi(c))^n \xrightarrow{n \rightarrow \infty} \min$$

$$c \leq 1 - \sqrt[n]{1 - \alpha}$$

$$\Phi(c) = \int_0^c \frac{e^{1-x}}{e-1} dx = \frac{e}{e-1} (1 - e^{-c})$$

$$\frac{d}{dc} \Phi(c) = \frac{e}{e-1} \cdot e^{-c} \downarrow \xrightarrow{\min} c = 1 - \sqrt[n]{1 - \alpha}$$

$$W = 1 - (1 - \Phi(c))^n = 1 - \left(1 - \frac{e}{e-1} (1 - e^{-c})\right)^n$$

$$1 - \sqrt[n]{1 - \alpha} = 1 - \left(1 - \frac{\alpha}{n} + o\left(\frac{1}{n}\right)\right) =$$

$$= \frac{\alpha}{n} + o\left(\frac{1}{n}\right)$$

$$e^{\frac{\alpha}{n} + o\left(\frac{1}{n}\right)} = 1 + \frac{\alpha}{n} + o\left(\frac{1}{n}\right)$$

$$W = \frac{e}{e-1} \cdot \left(1 - 1 + \frac{\alpha}{n} + o\left(\frac{1}{n}\right)\right) =$$

$$= \frac{e}{e-1} \cdot \frac{\alpha}{n} + o\left(\frac{1}{n}\right) \xrightarrow{n \rightarrow \infty} 0$$

$1_1) =$

T12

$H_0:$

$$P(4) = \frac{8}{24} = \frac{1}{3}$$

$$\underline{L = 0.2}$$

$$P(3) = \frac{1}{24} = \frac{1}{6}$$

H_1 : equal rooms

$$P(1) + P(2) = \frac{1}{2}$$

$$P(1) = P(2) = 1/4$$

$$L(x) = \frac{p(x)}{p(a)} = \frac{p(a,b)}{\frac{1}{16}} = 16 p(a,b) \geq C$$

$P_0:$

$$(1,1) = 1/16$$

$$(1,2) = 1/16$$

$$(1,3) = 1/24$$

$$(1,4) = 1/12$$

$$(2,1) = 1/16$$

$$(2,2) = 1/16$$

$$(2,3) = 1/24$$

$$(2,4) = 1/12$$

$$(3,1) = 1/24$$

$$(3,2) = 1/24$$

$$(3,3) = 1/36$$

$$(3,4) = 1/18$$

$$(4,1) = 1/12$$

$$(4,2) = 1/12$$

$$(4,3) = 1/18$$

$$(4,4) = 1/9$$

L:

$$1. 16/9 \quad (4,4)$$

$$2. 4/3 \quad (1,4), (2,4), (4,1), (4,2)$$

$$3. 1 \quad (1,1), (1,2), (2,1), (2,2)$$

$$4. 8/9 \quad (3,4), (4,3)$$

$$5. 2/3 \quad (1,3), (2,3), (3,1), (3,2)$$

$$6. 4/9 \quad (3,3)$$

$$P(L \geq c | H_0) \leq L = 0.2$$

~~H_1~~

G : forw. $(4,4)$

$$\alpha_1 = \frac{1}{9}$$

$$W = P(L \geq c | H_1) = \frac{1}{16}$$

$$\alpha_2 = 1 - W = \frac{15}{16}$$